COOLSHADE · DESIGN DECISION RECORD

CoolShade TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CSH-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/decisions/0001-trl2-review-decisions.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted. On 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos." Items D1 to D7 are "Decided by Amish, 2026-09-25: go with recommendation". O1 (budget) is decided at the latest recommendation, $775 (see CSH-DDR-002). O2 has no recommendation and remains "Proposed, awaiting Amish".

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", eight of them with a recommendation, and the design precis CSH-PRC-001 v0.2 listed six key design choices, all proposed. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the CoolShade items one by one. Under that instruction, every item with a recommendation is adopted as recommended so that TRL 3 work can proceed, and each stays open for his review. Items without a recommendation stay open. A recommended budget change is not applied to `project.yaml`; it is recorded here as awaiting Amish.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (session 2026-09-25, /populate) and in CSH-PRC-001 v0.2, Key design choices.

Decision

Table 1. Items decided.

#	Item	Recommendation	Status
D1	Misting pressure class	Low pressure, about 7 bar, 12 V, off-grid; droplet size and wetting are the first things to check	Decided by Amish, 2026-09-25: go with recommendation
D2	Activation thresholds	Fixed thresholds: 32 °C or more, 60 % relative humidity or less, presence in the last 2 min, tank above its low level	Decided by Amish, 2026-09-25: go with recommendation
D3	Fans	None at TRL 3; revisit because R4 remains unmet (see `docs/REVIEW.md`)	Decided by Amish, 2026-09-25: go with recommendation
D4	Water supply	Tank as standard; mains connection with a float valve and backflow preventer optional where local rules allow	Decided by Amish, 2026-09-25: go with recommendation
D5	Canopy size and structure	3.0 x 2.4 m, four 80 x 80 x 3 mm posts with knee braces, bolted to an existing slab; checked by calculation in CSH-CAL-001	Decided by Amish, 2026-09-25: go with recommendation
D6	Storm practice	Removable fabric, taken off before forecast storms. CSH-CAL-001 shows this is structurally necessary, not only a convenience	Decided by Amish, 2026-09-25: go with recommendation
D7	Status link	None for the first prototype; no FieldNode link fitted	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that were open after TRL 3.

#	Item	Status
O1	Budget. The TRL 2 recommendation was option (a), raise `budget_usd` to $750. Under the batch instruction, budget figures are not changed: `budget_usd` stays at $700 and the $750 figure is recorded here as a recommendation. CSH-CAL-001 v0.1 found a parts cost of $758, over both figures, and the TRL 3 review recommended $775.	Decided by Amish, 2026-09-25: go with recommendation ($775; see CSH-DDR-002)
O2	First site type and co-design partner (bus stop, market or queue). No recommendation was made.	Proposed, awaiting Amish

Consequences

- `project.yaml` : only the TRL fields change. `budget_usd` stays at $700. No pitch or problem rewording was recommended, so none is applied.
- CSH-PRB-001, CSH-PRC-001 and CSH-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed"; they are adopted for TRL 3 pending Amish's review. No requirement target changes; the R6 thresholds are now adopted rather than proposed, and R13 is stated against both $700 and the proposed $750.
- The TRL 3 calculations led to four detail changes within these decisions: shorter knee braces (headroom), a vacuum breaker on the mist line (draining past anti-drip nozzles), a closed mist line loop, and a tank strap (an empty tank tips in a 30 m/s gust). They add $23 to the parts cost.
- CSH-CAL-001 finds R4, R10 and R13 not met. New options for R10 (fabric-on wind limit or heavier roof beams) and a revisit of fans for R4 are listed in `docs/REVIEW.md` as proposed, awaiting Amish; this record does not decide them.
- Update, 2026-09-25: Amish accepted all recommendations. D1 to D7 and O1 are decided; the R10, R4 and R8 items and their effects are recorded in CSH-DDR-002. O2 stays open.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.